ModelIt!

Teacher's Guide

Why Earthquakes Hit Some Cities Harder

Modeling Seismic Energy, Ground Conditions, and Urban Vulnerability

Quick Reference

Grade Level	Grade 09L1
NGSS Standard	HS-ESS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-1

Develop a model to illustrate how Earth's internal and surface processes operate at different spatial and temporal scales to form continental and ocean-floor features.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model to illustrate how tectonic processes at depth interact with surface conditions to produce variable earthquake impacts across spatial scales.
Disciplinary Core Idea	**ESS2.B: Plate Tectonics and Large-Scale System Interactions** — Earth's internal processes (tectonic stress, fault mechanics) operate at temporal scales of centuries and spatial scales of thousands of kilometers, while surface effects (ground amplification, building response) operate at human timescales and local spatial scales.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students analyze how earthquake phenomena span dramatically different scales: tectonic stress over centuries, fault slip in seconds, seismic waves over minutes, and ground amplification over meters — all interacting to determine outcomes.

CAST Testing Alignment

MULTIPLE CHOICE:

Two neighborhoods are exactly 10 km from an earthquake epicenter. Neighborhood A is built on granite bedrock; Neighborhood B is built on water-saturated clay fill. According to your model, which will experience more intense shaking, and why?

CONSTRUCTED RESPONSE:

The 2010 Haiti earthquake (magnitude 7.0) killed 230,000 people, while the 2010 Chile earthquake (magnitude 8.8 — nearly 500 times more energy) killed 525 people. Using your model, explain how two earthquakes of dramatically different power could produce such opposite death tolls.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Earthquakes are among the most dramatic reminders that Earth's surface is not the stable, unchanging ground it appears to be. Beneath our feet, tectonic plates grind against each other at rates of centimeters per year, building enormous stress that is released in catastrophic bursts lasting seconds to minutes. But the earthquake itself is only half the story — what happens to seismic energy when it reaches the surface determines whether a city survives or collapses.

Tectonic Stress and Fault Mechanics

Earth's tectonic plates move at 1-15 cm per year, driven by mantle convection. Where plates meet, friction locks them together, and stress accumulates along the boundary. When stress exceeds the fault's frictional strength, the rocks suddenly snap past each other — releasing decades to centuries of stored energy in seconds. The San Andreas Fault, for example, has accumulated about 5 meters of 'slip deficit' since its last major rupture in 1906 — enough for a magnitude 7.8+ earthquake.

Seismic Wave Propagation

An earthquake radiates three types of waves: P-waves (compressional, fastest, arrive first), S-waves (shear, slower, more damaging), and surface waves (slowest, most destructive). These waves travel through Earth's interior and along its surface at speeds of 3-8 km/s. As they encounter different rock and soil types, their behavior changes dramatically — they can be reflected, refracted, amplified, or dampened depending on the medium's density and elasticity.

Ground Amplification: The Site Effect

This is the single most important factor in determining local earthquake damage. When seismic waves travel from dense bedrock into loose, soft sediments, conservation of energy forces the waves to slow down and increase in amplitude — like ocean waves growing taller as they approach the shore. Soft clay can amplify shaking 2-10 times compared to bedrock. The 1985 Mexico City earthquake killed 10,000 people largely because the city is built on an ancient lakebed that amplified seismic waves like a bowl of gelatin, even though the earthquake's epicenter was 350 km away.

Building Vulnerability and Engineering

Modern earthquake engineering uses base isolation (buildings float on rubber bearings), moment-resisting frames (flexible steel that absorbs energy), shear walls (concrete cores that resist lateral forces), and tuned mass dampers (counterweights that reduce sway). The difference is stark: in the 2010 Chile earthquake, modern buildings designed to seismic codes suffered minimal damage while unreinforced masonry structures nearby were destroyed. Building codes save lives — but only if they exist, are enforced, and buildings are actually constructed to meet them.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify tectonic stress, fault slip, seismic wave energy, building vulnerability, and ground amplification as the key components that determine earthquake impact.

E **Establish Relationships**
Students determine that tectonic stress drives fault slip, which generates seismic wave energy, which is then modified by ground amplification and interacts with building vulnerability to determine destruction levels.

V **Visualize & Model**
Students build a computational model showing how geological processes at depth cascade through surface conditions to produce variable earthquake impacts across a city.

E **Evaluate Outcomes**
Students run bedrock, soft soil, and two-city comparison scenarios to test their model and discover that local conditions often matter more than earthquake magnitude.

R **Revise & Extend**
Students add aftershock frequency, tsunami potential, or infrastructure interdependency to model more realistic post-earthquake cascading effects.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In 2010, Haiti had a magnitude 7.0 earthquake that killed 230,000 people. Two months later, Chile had a magnitude 8.8 — that's 500 times more energy — and 525 people died. How is that possible?"*

Watch For

Show the comparison on screen. Let the numbers sink in. Ask students what could possibly explain this.

Slide 2: Learning Objectives

3 min

Say: *"Today you're learning why earthquakes are only half the story — the ground you're standing on and the building you're in determine whether you live or die."*

Watch For

Have students read objectives. Pre-teach 'ground amplification' and 'liquefaction.' Use the gelatin bowl analogy.

Slide 3: Big Question

3 min

Say: *"Same earthquake, same distance. One city stands. One city falls. The earthquake didn't choose — the geology and engineering did."*

Watch For

Show a map with two cities equidistant from a fault. Ask students to predict which would fare worse and why.

Slide 4: LEVER Framework

2 min

Say: *"We're building a model that starts deep underground where stress is building and follows the energy all the way up to the buildings above."*

Watch For

Preview LEVER steps. Emphasize that this model spans scales from tectonic plates to individual buildings.

Slide 5: Activity 1: Components

8 min

Say: *"Only one component is truly 'external' — the one we can't control. What is it, and why can't we control it?"*

Watch For

Guide sorting: tectonic stress is external (geological force we can't influence). All other components are internal system responses. Discuss implications.

Slide 6: Activity 2: Connections

8 min

Say: *"Stress builds for 100 years. It releases in 10 seconds. Then what happens to the energy between the fault and your school?"*

Watch For

Help students trace the energy path: stress → slip → waves → amplification → damage. Identify where intervention is possible.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's put two cities through the same earthquake and see which one survives. What's the difference?"*

Watch For

Students predict outcomes. Run all three scenarios. Key discovery: ground amplification can matter more than earthquake magnitude.

Slide 8: Discoveries

5 min

Say: *"The deadliest earthquakes aren't the strongest. They're the ones that hit unprepared cities on bad soil. That's both terrifying and hopeful — because it means we CAN do something about it."*

Watch For

Lead discussion on Haiti vs. Chile. Connect preparedness to equity: who builds on the worst soil? Usually the poorest communities.

Slide 9: STEM Challenge

12 min

Say: *"Your city hasn't had a big earthquake in 120 years. That's not comforting — that means MORE stress has built up. Design a system that identifies who's most at risk."*

Watch For

Groups create risk assessment maps and early warning systems. Must consider soil type, building quality, and community communication.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Tectonic Stress
- **INTERNAL:** Fault Slip, Seismic Wave Energy, Building Vulnerability, Ground Amplification

Relationships

- **Tectonic Stress to Fault Slip:** POSITIVE (+) — Greater accumulated tectonic stress produces larger fault slip when the rocks finally break. More stored energy means a longer rupture, greater displacement, and a higher magnitude earthquake.
- **Fault Slip to Seismic Wave Energy:** POSITIVE (+) — Larger fault slip releases more energy as seismic waves. A 1-meter slip produces moderate shaking; a 10-meter slip (like the 2011 Japan earthquake) releases catastrophic energy across vast distances.
- **Seismic Wave Energy to Ground Amplification:** POSITIVE (+) — Stronger seismic waves traveling through soft sediments produce greater amplification effects. The interaction is multiplicative — double the wave energy in soil that amplifies 5x results in 10x the shaking compared to bedrock.
- **Ground Amplification to Building Vulnerability:** POSITIVE (+) — Greater ground amplification increases the seismic forces that buildings must withstand. Buildings designed for moderate shaking may survive on bedrock but fail catastrophically when ground amplification doubles or triples the effective shaking intensity.

Key Discoveries

- Ground amplification is often more important than distance from the earthquake — soft soil can amplify shaking 2-10 times, meaning a city on clay can experience worse shaking than a closer city on bedrock
- Tectonic stress builds silently over decades to centuries, then releases in seconds — the longer the buildup, the more powerful the potential earthquake
- Building vulnerability varies enormously with construction quality — modern earthquake-resistant buildings can survive shaking that destroys unreinforced structures just blocks away
- The deadliest earthquakes aren't necessarily the strongest — a moderate earthquake in a vulnerable city can kill more people than a powerful earthquake in a prepared one

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Earthquakes kill people"	Earthquakes don't kill people — buildings kill people. The ground shaking itself is rarely directly lethal. Casualties result from collapsing structures, falling debris, fires from ruptured gas lines, and tsunamis. In well-engineered cities, even powerful earthquakes cause few deaths. The 2010 Chile magnitude 8.8 earthquake killed 525 people because of strict building codes; the 2010 Haiti magnitude 7.0 killed 230,000 because of absent building codes.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All earthquakes produce the same type of shaking"	Different earthquakes produce different frequency content depending on fault mechanics and depth. Smaller, shallow earthquakes produce high-frequency shaking that affects short, stiff buildings. Larger, deeper earthquakes produce low-frequency waves that cause tall buildings to sway dangerously. Ground conditions further modify the frequency content — soft soil amplifies low-frequency waves that are particularly damaging to mid-rise buildings.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We can predict when earthquakes will happen"	Despite decades of research, reliable short-term earthquake prediction remains impossible. We can identify WHERE earthquakes are likely (plate boundaries) and roughly HOW STRONG they might be (based on fault length and accumulated stress), but we cannot predict WHEN the fault will slip. The best approach is not prediction but preparation — building codes, early warning systems, and emergency plans that assume an earthquake can happen any time.